



Flood predictor

QGIS Plugin: User Guide

Open-source flood inundation modelling for QGIS

Dr. Mohammed Hlal

GeoAI & Disaster Risk Management

mohammed.hlal@ump.ac.ma

Plugin version 1.2.1

July 2026

Table of Contents

1. Introduction	3
2. Requirements.....	3
3. Installing the Plugin	4
4. Interface Overview	5
5. Step-by-Step Workflow	5
6. Interpreting the Results	7
7. Tips & Troubleshooting.....	8
8. Support.....	9

1. Introduction

Flood Predictor is a QGIS plugin that estimates flood-inundation extent and depth for a chosen area, without requiring a hydraulic-modelling background or specialist input datasets. It automatically downloads open elevation data, identifies existing water bodies in your imagery, and grows a flood surface outward from them to produce ready-to-use GIS layers.

1.1 What the plugin does

- Automatically fetches a Digital Elevation Model (DEM) for your study area no need to source or clip your own elevation raster.
- Repairs the DEM (“depression-fill”) so small pits or mosaic artefacts don't break the flood-flow logic.
- Identifies existing water pixels (“seeds”) from your imagery using the NDWI water index, or from a water mask/vector layer you already have.
- Grows the flood extent outward from those seed pixels up to a target water level, following a connectivity rule so that low-lying but unconnected land is correctly left dry.
- Exports the result as GeoTIFF (extent and/or depth) and/or Shapefile, for one or several flood-depth scenarios at once.

1.2 What kind of model this is

Flood Predictor uses a simplified, geometric (“bathtub”) approach constrained by hydraulic connectivity. It is well suited to fast, first-pass flood susceptibility screening, teaching, and preliminary risk mapping. It is not a hydrodynamic model: it does not simulate flow routing, discharge, or how flooding evolves over time (the way HEC-RAS or LISFLOOD-FP would). Treat its outputs as a susceptibility screening layer, not an engineering-grade forecast.

2. Requirements

- QGIS 3.16 or later.
- An active internet connection (the plugin downloads DEM tiles from a public AWS dataset no account or API key needed).
- Reference imagery or a water layer for your study area, to identify existing water bodies. Any one of the following works:
 - Optical Green and Near-Infrared bands (e.g. Sentinel-2 bands B03 and B08, or Landsat equivalents) for NDWI-based seeding
 - An existing water/flood raster mask
 - An existing river, lake, or water polygon/line vector layer

No GDAL, numpy, or other add-ons need to be installed separately the plugin only relies on components already bundled with QGIS.

3. Installing the Plugin

Flood Predictor is installed like any other QGIS plugin, from the distributed .zip file:

1. In QGIS, open the Plugins menu (Extensions, in a French-language install) and choose Manage and Install Plugins.
2. Switch to the Install from ZIP tab.
3. Click Browse, select flood_predictor.zip, then click Install Plugin.
4. Make sure the checkbox next to “Flood Predictor” is ticked in the plugin list, if it isn't enabled automatically.
5. A new toolbar icon appears, and the tool is also available from Plugins → Flood Predictor.

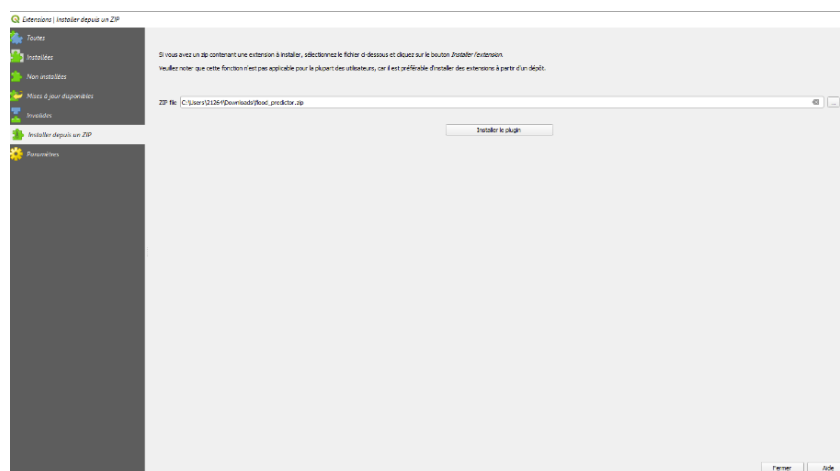


Fig. 1 Accessing the plugin manager from the Extensions (Plugins) menu.

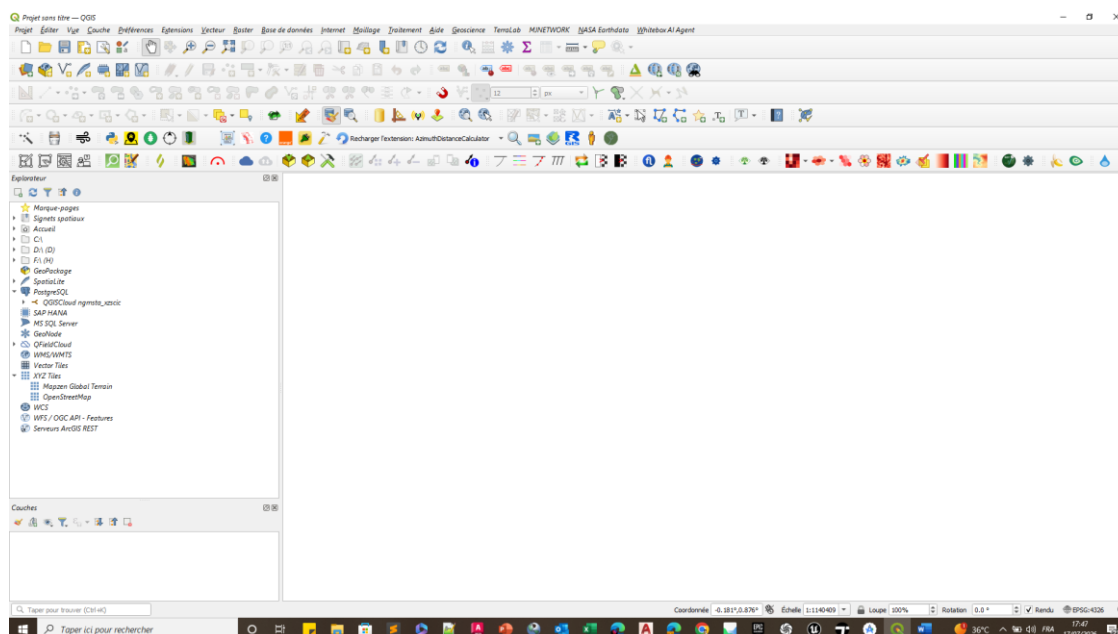


Fig. 2 Installing from a ZIP file: browse to flood_predictor.zip and click Install Plugin.

4. Interface Overview

The Flood Predictor dialog is organised into four groups, filled in from top to bottom:

- Study Area the bounding box the DEM will be downloaded for, plus the DEM resolution (zoom level).
- Water Seeding how the plugin should identify existing water pixels (NDWI, a water raster, or a water vector layer).
- Inundation Model the flood depth(s) to simulate and the model's technical parameters.
- Outputs which result layers to export, and where to save them.

Before opening the tool, load your reference imagery (or existing water layer) into QGIS as you normally would, so it is available in the layer dropdowns.

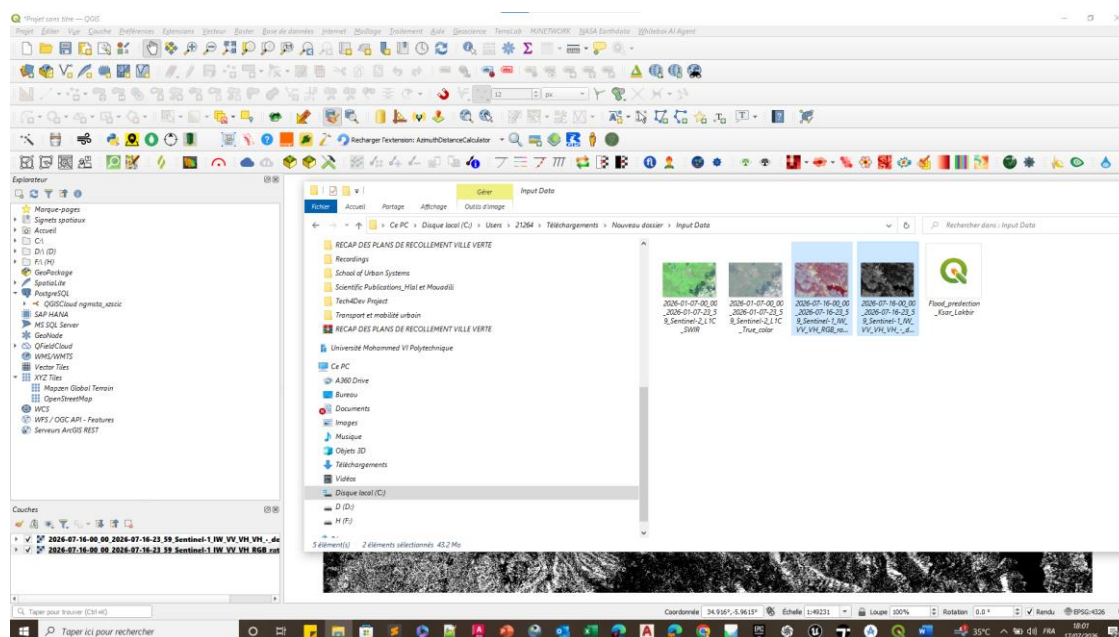


Fig. 3 Reference imagery imported into QGIS, ready to be used for water seeding.

5. Step-by-Step Workflow

5.1 Define the study area

- Open the Flood Predictor tool from the toolbar or Plugin's menu.
- Click Use Current Canvas Extent to take the bounding box directly from your current map view (it is converted to WGS84 automatically), or type coordinates in manually.
- Set the DEM zoom level. 12 is a good default (roughly 20–40 m per pixel); higher values give finer resolution but download more tiles and run slower. The live tile-count estimate helps you judge this request over 400 tiles are rejected, so shrink the area or lower the zoom if you hit that limit.

5.2 Choose a water-seeding method

Pick the option that matches the imagery you have available:

- NDWI from Green/NIR raster bands select a Green-band raster and a NIR-band raster, and set the NDWI threshold (0.0 is the standard cut-off; a slightly negative value such as -0.1 can help pick up narrow or shallow channels).
- Existing water raster mask select a classified raster and the pixel value that represents water.
- Existing water vector layer select a river/lake layer (points, lines, or polygons); it is rasterised onto the DEM grid automatically.

Important NDWI needs true optical bands

The Normalized Difference Water Index is defined as $(\text{Green} - \text{NIR}) / (\text{Green} + \text{NIR})$ and requires genuine optical reflectance bands, such as Sentinel-2 B03 (Green) and B08 (NIR).

Sentinel-1 radar (SAR) products including VV/VH backscatter ratios or decibel-gamma0 layers are not Green or NIR bands and feeding them into the Green/NIR fields will give a meaningless NDWI and an unreliable seed mask.

If your only imagery is SAR-based, first derive a water mask from it in QGIS (e.g. an amplitude threshold) and load that as an Existing water raster mask instead of using the NDWI mode.

5.3 Set the inundation model parameters

- Flood depth(s) (m) one value, or a comma-separated list for several scenarios in a single run, e.g. 0.5,1,2.
- Connectivity 8 (orthogonal + diagonal) is the default and usually the better choice; 4 (orthogonal only) is stricter about what counts as a connected path.
- Seed reference level how the reference water elevation is summarised from the seed pixels: median (default, robust to outliers), min, or mean.
- Depression-fill preprocessing leave this ticked (recommended); it removes small DEM pits that would otherwise wrongly block or isolate the flood fill.

5.4 Configure outputs and run

- Tick and set a save path for any combination of: flood extent raster (GeoTIFF), flood depth raster (GeoTIFF), and flood extent vector (Shapefile).
- With several flood depths, each output is written per depth (suffixed _<depth>m), alongside a combined staging raster showing the minimum depth at which each cell becomes inundated.
- Click Run. Processing happens in the background QGIS stays usable with progress and log messages shown at the bottom of the dialog. Click Cancel at any time to stop it.
- When it finishes, the result layers are added to your project automatically.

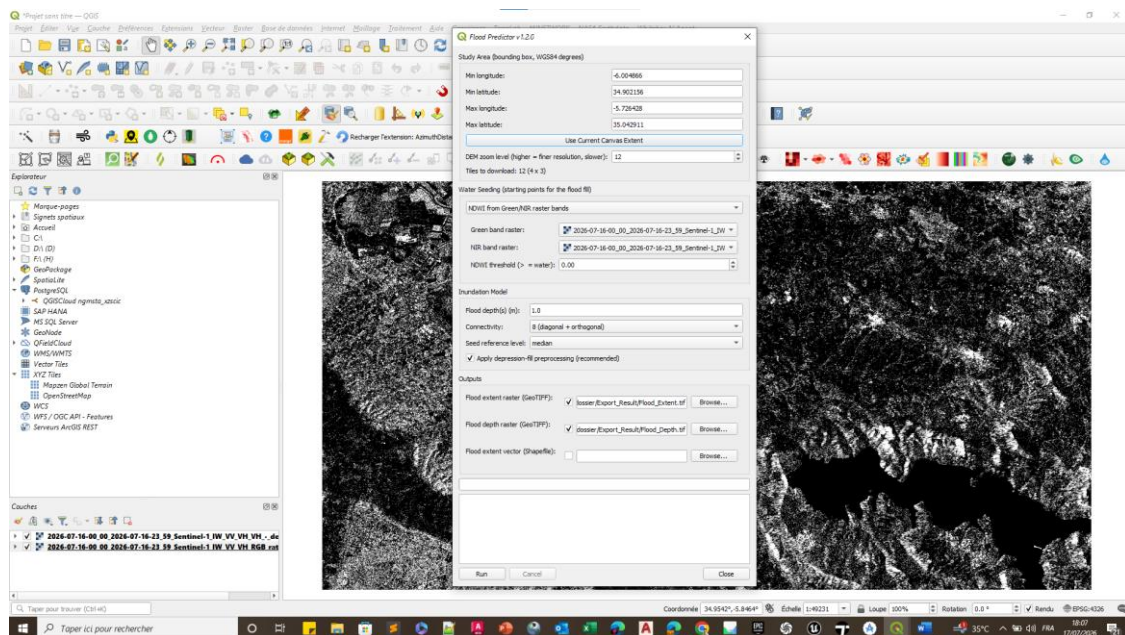


Fig. 4 The Flood Predictor dialog fully configured: study area, water seeding, model parameters, and outputs.

6. Interpreting the Results

Once the run completes, the exported layers are added to the map so you can inspect them straight away:

- The flood extent layer marks cells predicted to be inundated at the chosen depth(s) style it as a semi-transparent fill so the base map remains visible underneath.
- The flood depth layer gives the estimated water depth (in metres) at each inundated cell; a graduated colour ramp (e.g. light blue → dark blue) works well for showing depth variation.
- With multiple depth scenarios, compare layers side by side to see how the flooded area grows between scenarios, which is useful for identifying the areas most exposed even at low flood depths.

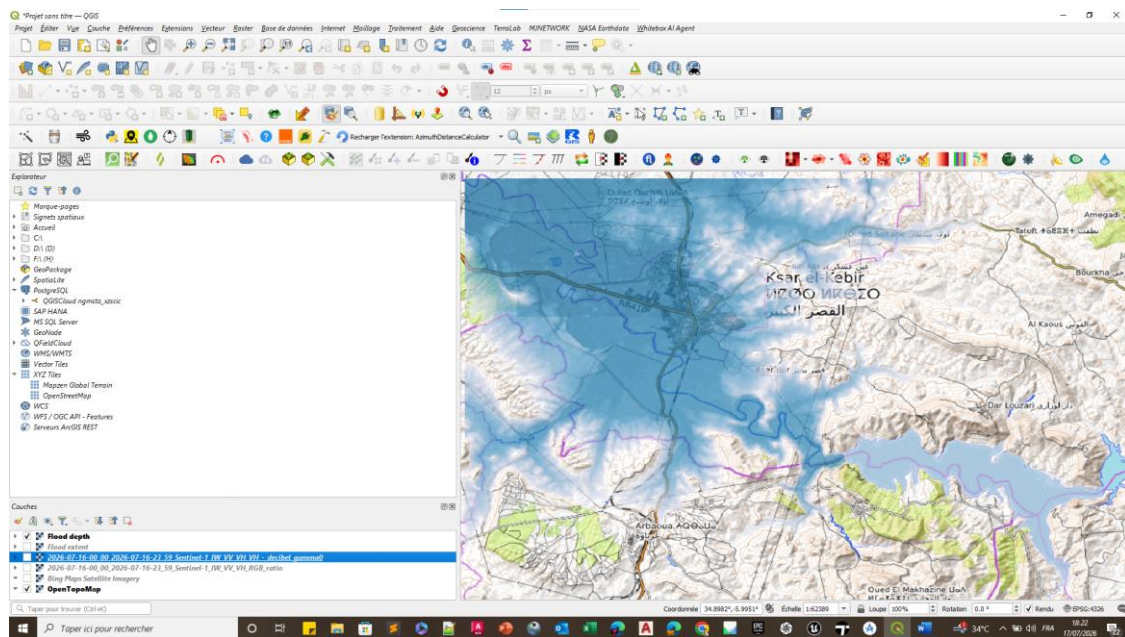


Fig. 5 Predicted flood extent displayed over the study area.

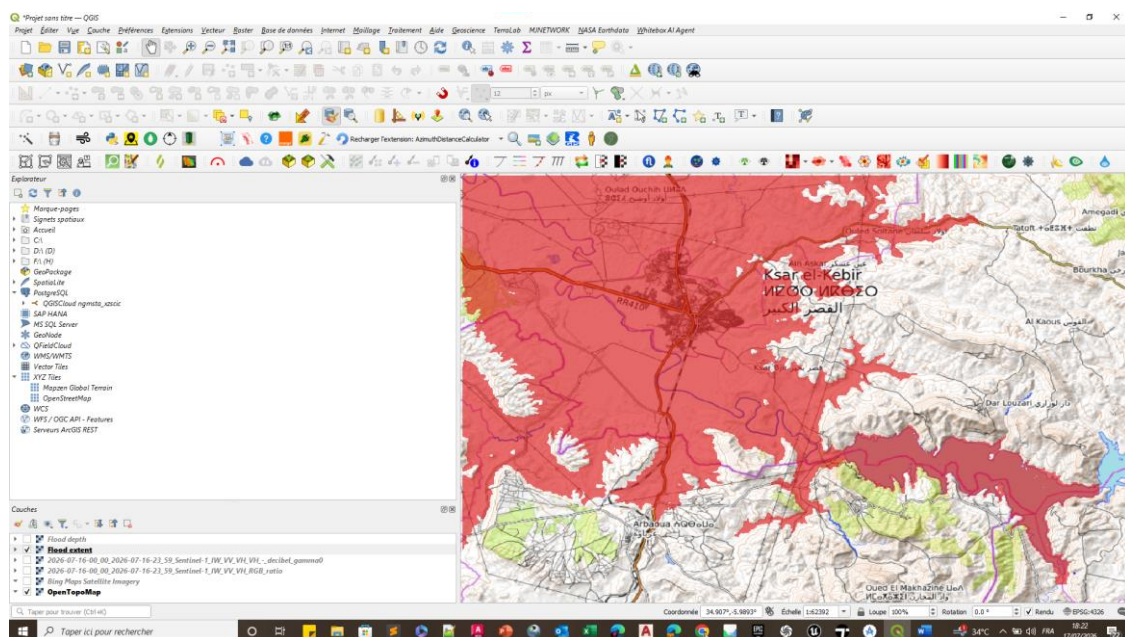


Fig. 6 Flood depth output, showing estimated water depth across the inundated area.

7. Tips & Troubleshooting

“Tiles to download” exceeds the limit

Requests over 400 DEM tiles are rejected to avoid very slow runs. Shrink the study area, lower the DEM zoom level, or split the area into smaller tiles processed separately.

A layer doesn't appear in the Green/NIR/water dropdowns

Inputs must be file-based raster layers (e.g. a GeoTIFF) that QGIS can trace back to a file path some in-memory or virtual/temporary layers are not currently supported. Save the layer to disk first, then reload it.

The flood extent looks patchy or misses obvious low ground

Make sure Depression-fill preprocessing is enabled, and check that your seed pixels are actually connected (by land, hydraulically) to the area you expect to flood the connectivity constraint deliberately excludes low-lying but disconnected depressions.

Precise area or volume calculations

The DEM mosaic is georeferenced directly from slippy-map tile coordinates in WGS84 degrees, which is standard for tile-based DEM tools but introduces small distortion for area/volume math. For precision work, reproject the exported layers into a suitable projected CRS (e.g. a local UTM zone) before measuring.

Remember the model's scope

Flood Predictor is a planar/"bathtub" connectivity model, not a hydrodynamic solver. Use it for rapid susceptibility screening and communication, and pair it with a full hydraulic model (e.g. HEC-RAS, LISFLOOD-FP) wherever an engineering-grade result is required.

8. Support

Flood Predictor version 1.2.1

Author: Dr. Mohammed Hlal GeoAI & Disaster Risk Management

Email: mohammed.hlal@ump.ac.ma

For bug reports or feature requests, please use the plugin's issue tracker on its GitHub repository.